

WEAVE: Escaping the Identity Shortcut in Lightweight Style Transfer with Wavelets

Anonymous ICME submission

Abstract—Lightweight style transfer suits consumer devices, yet many compact models return an image closer to the unchanged input than to the requested style, and standard metrics reward this identity shortcut. We make the failure measurable with two trivial references, the unchanged source and an unrelated target-style image, which bracket valid transfer. On three benchmarks, eight of thirteen recent methods leave this bracket at least once. We trace the shortcut to frequency dominance: coarse structure carries most of the training gradient but little style information. WEAVE resolves it in the latent space of a frozen autoencoder. A Haar transform separates structure from texture, a compact velocity network learns source-anchored transport, and stepwise statistical alignment injects high-frequency style. With 1.04M trainable parameters and 1.4 minutes of training on one consumer GPU, WEAVE stays inside the bracket on all benchmarks and preserves the most content among methods that do.

Index Terms—style transfer, flow matching, wavelet decomposition, evaluation protocol, efficient generation

I. INTRODUCTION

Artistic style transfer has moved from research demos to camera apps, creative tools, and on-device media pipelines, where memory and latency budgets are tight. The strongest current results come from diffusion editors and image foundation models, which reuse a billion-parameter denoiser at every call and can require more than 60 GB of GPU memory [1], [2]. Compact feed-forward and state-space models [3]–[5] fit these budgets, but they raise a question that current benchmarks rarely ask: *did the output actually change style?*

A hidden failure mode. Content metrics such as LPIPS [6] and the content factor of ArtFID [7] are maximized by copying the input, so a model that barely edits the source can look excellent. We therefore report two trivial outputs with every metric (Fig. 1): Identity (IDT) returns the source, and Target Style (TGT) returns an unrelated image of the requested style. Valid transfer must beat IDT on style and stay closer to the source than TGT on content. This *IDT–TGT sandwich* is simple, uses raw metric units, and is revealing: across three benchmarks, eight of thirteen recent methods fall outside it at least once (Table I). Some are indistinguishable from a copy in style, which we call the *identity shortcut*; others discard as much content as an unrelated painting.

Why compact models take the shortcut. We find the cause in the training signal rather than in model size. When a velocity field is regressed on the full latent, the low-frequency (LL) Haar band carries 69.5% of the gradient energy across 5,000 cross-style pairs, yet it is the *least* style-discriminative band: its between-/within-style variance ratio is 0.12, against 0.56 for the finest band HH (Fig. 2). The objective is thus dominated by

structure that should change little, while the texture that defines style receives a weak, easily ignored gradient. Networks that fit low frequencies first [8], [9] converge toward the source, and widening them does not help because both signals still share one basis and one loss.

Our approach. WEAVE decides *where* each signal lives before deciding how much capacity to spend. Working on cached latents of a frozen SD1.5 VAE [10], a one-level Haar transform separates structure (LL) from oriented texture (LH/HL/HH). A 1.04M-parameter velocity network learns transport toward a source-anchored target under a down-weighted LL loss; pooled, coordinate-free target codes steer only the LH/HL heads; and an AdaIN update aligns the high-frequency residual after every Euler step. Training takes 82.8 s on one RTX 3060 and stops at an internal gradient transition, without any image evaluation.

WEAVE is the only learned model under 10M parameters that stays inside the sandwich on all three benchmarks, and among the five methods that do, it preserves the most content, with the highest DINO-C on every benchmark, 11–34× fewer trainable parameters than the other learned ones, and 11–18× faster inference than the others with a reported runtime.

Contributions. (1) *Evaluation.* The IDT–TGT sandwich, a metric-agnostic protocol that exposes the identity shortcut and content collapse, and an audit of thirteen methods under it. (2) *Diagnosis.* A subband measurement showing that low-frequency gradient dominance, not capacity, drives compact latent models toward identity. (3) *Method.* Wavelet-decoupled velocity matching that assigns structure, learned texture motion, and trajectory-level statistics to explicit subbands, with metric-free early stopping. (4) *Evidence.* The best content preservation among valid methods at 1.04M parameters and 1.4 training minutes, supported by matched ablations, three seeds, reference-pool resampling, zero-shot transfer, and a training-free plug-in that improves a frozen SD1.5 editor.

II. RELATED WORK

Neural and statistical style transfer. Gatys et al. matched Gram statistics by optimization [11], [12]; AdaIN and WCT replaced it with feed-forward moment and covariance transforms [13], [14]. SANet, StyTR-2, and AesPA-Net add attention [15]–[17], CUT learns unpaired translation by patch contrast [5], and Mamba-based SaMST and SaMam reduce sequence cost [3], [4], [18]. These methods mix structure and appearance in one feature tensor; WEAVE restricts statistical correction to explicit frequency coordinates.

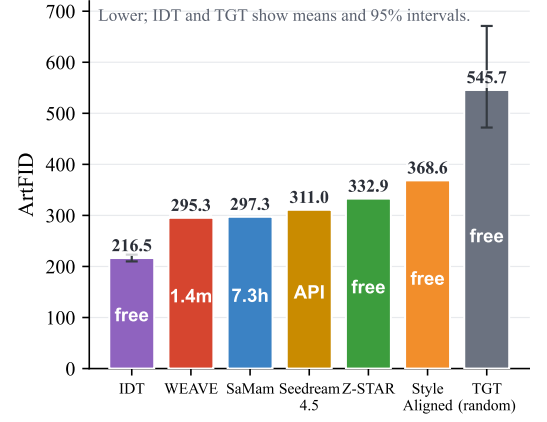
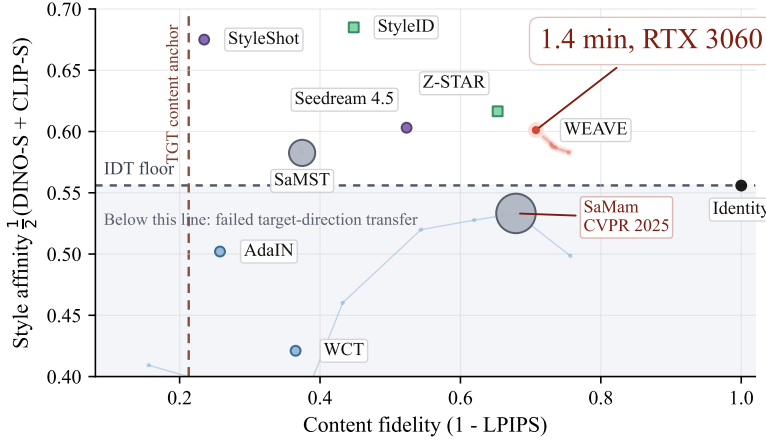


Fig. 1. IDT-TGT sandwich on D5-512. Left: style affinity versus content fidelity; bubble area is training time. Methods below the IDT line have not moved toward the target style; methods left of the TGT line have lost as much content as an unrelated target-style image. Right: target-pooled ArtFID, where IDT scores best, showing why a no-op reference is needed (IDT: 95% bootstrap interval; TGT: 2.5–97.5 percentile over 30 reference draws).

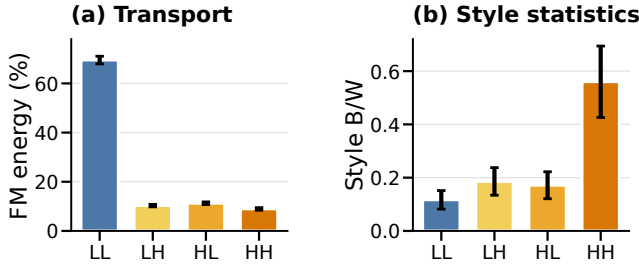


Fig. 2. Frequency probes on 5,000 cross-style latent pairs. (a) Share of endpoint-displacement energy per Haar band. (b) Between-/within-style variance ratio of per-image band statistics. Transport energy concentrates in LL while style separation is strongest at high frequency.

Diffusion editors. Latent diffusion [10] enables SDEdit, Prompt-to-Prompt, StyleID, StyleAligned, InST, Z-STAR, IP-Adapter, and StyleShot [19]–[26]. Being “training-free” leaves a 1B-parameter denoiser, often with inversion, in every call. WEAVE keeps only the frozen VAE; our plug-in experiment tests the Haar operator inside such an editor.

Flow matching and bridges. Rectified flow and flow matching [27], [28] regress velocities along endpoint interpolations, and Schrödinger-bridge solvers [29]–[32] address unpaired domains, but their objectives weigh all coordinates together. WEAVE changes the endpoint, basis, and per-band weights of the same regression.

Frequency-aware and wavelet methods. FDIT constrains low and high frequencies separately for translation [33], FreqFlow weights frequencies for class-conditional generation [34], and wavelets support diffusion [35] and photorealistic transfer [36]. WEAVE uses the orthogonal Haar basis [37] *inside* the transport objective, with a distinct role per band.

III. METHOD

A. Latent Haar Coordinates

Let $z_c = \mathcal{E}(x_c) \in \mathbb{R}^{4 \times H \times W}$ and $z_s = \mathcal{E}(x_s)$ be content and target-style latents of the frozen VAE. The one-level Haar transform $\mathcal{W}(z) = (\ell, h_1, h_2, h_3)$ is orthonormal and exactly invertible, so losses and heads can be assigned to explicit coordinates. We write $P_L z = \mathcal{W}^{-1}(\ell, 0, 0, 0)$ and $P_H = I - P_L$; latents with equal $P_L z$ share the LL coordinate but may differ in local appearance. Across 5,000 cross-style pairs, displacement energy concentrates in LL while style separability is largest in HH (Fig. 2): the band that dominates the regression is not the band that carries style. The separation is exact in latent space, although the nonlinear decoder can still couple visible structure and appearance.

B. Source-aligned Endpoint

A direct path from z_c to z_s also learns the unrelated geometry of x_s . WEAVE instead keeps the source LL as an anchor and takes high frequencies from the target:

$$z_1^* = \mathcal{W}^{-1}((1-\alpha)\ell_c + \alpha \text{AdaIN}(\ell_c, \ell_s), h_{1,s}, h_{2,s}, h_{3,s}), \quad (1)$$

with $\alpha = 0.3$, which permits a limited palette shift. Training samples a lightly perturbed interpolation

$$z_t = (1-t)z_c + tz_1^* - \sigma\sqrt{t(1-t)}\epsilon, \quad u = z_1^* - z_c, \quad (2)$$

with $t \sim \mathcal{U}[0, 1]$, $\epsilon \sim \mathcal{N}(0, I)$, and $\sigma = 0.02$. The perturbation vanishes at both endpoints and is absent at inference; the label remains the clean displacement.

C. Band-weighted Velocity Matching

The network $v_\theta(z_t, t, s)$ predicts $(v_\ell, v_{h_1}, v_{h_2})$ with separate 1×1 heads on a shared trunk of four width-64 residual blocks. With $(u_\ell, u_{h_1}, u_{h_2}, u_{h_3}) = \mathcal{W}(u)$,

$$\mathcal{L} = \mathbb{E}_t [\lambda_{LL} \|v_\ell - u_\ell\|^2 + \|v_{h_1} - u_{h_1}\|^2 + \|v_{h_2} - u_{h_2}\|^2], \quad (3)$$

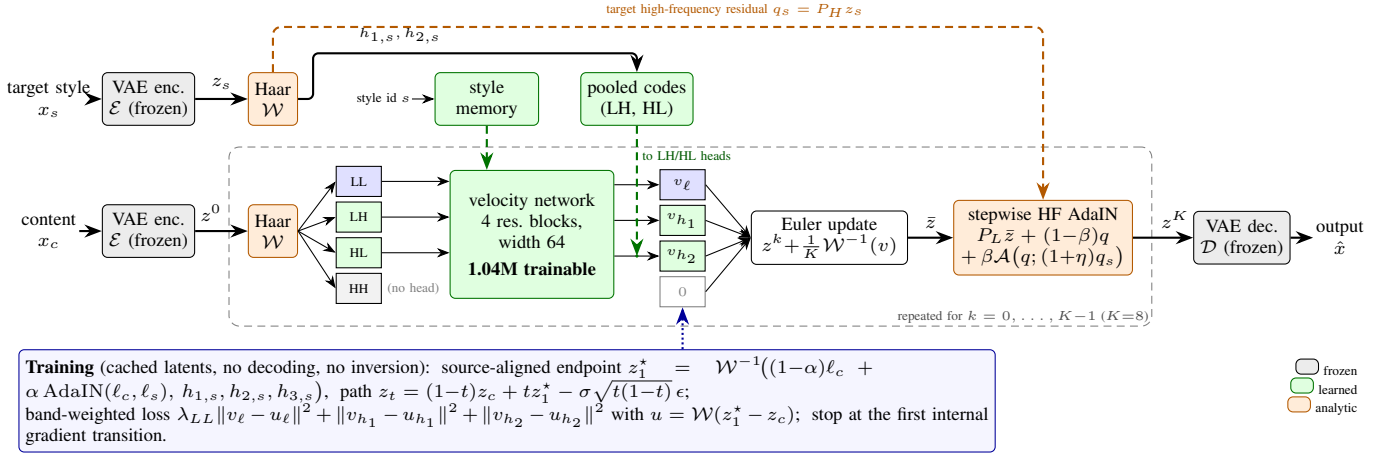


Fig. 3. WEAVE. A one-level Haar transform exposes LL and three high-frequency bands of the VAE latent. The compact network predicts LL/LH/HL velocities; pooled target LH/HL codes modulate only the matching heads and HH has no learned head. After every Euler update, AdaIN aligns the combined high-frequency residual to the target statistics. Only the green blocks are trained.

with $\lambda_{LL} = 0.3$. Changing the endpoint (Eq. 1) alters the *requested* velocity, whereas λ_{LL} alters its optimization pressure. A style memory indexed by the target-style id gives a coarse prior to the trunk. For image-specific texture, the target LH and HL bands are pooled into coordinate-free, orientation-specific codes that enter only the matching heads; raw spatial target maps would leak target layout. HH has no learned head. The complete model has 1,037,087 trainable parameters.

D. Stepwise High-frequency Alignment

Velocity matching organizes LL/LH/HL motion but leaves a weak appearance signal. For a residual $q = P_H z$ and target residual $q_s = P_H z_s$, let $\mathcal{A}(q; q_s) = \sigma(q_s)(q - \mu(q))/\sigma(q) + \mu(q_s)$ with channel-wise spatial statistics [13]. One of K inference steps is

$$\begin{aligned} \bar{z}^{k+1} &= z^k + \frac{1}{K} \mathcal{W}^{-1}(v_\ell^k, v_{h_1}^k, v_{h_2}^k, 0), \\ z^{k+1} &= P_L \bar{z}^{k+1} + (1 - \beta) q^{k+1} + \beta \mathcal{A}(q^{k+1}; (1 + \eta) q_s), \end{aligned} \quad (4)$$

with $q^{k+1} = P_H \bar{z}^{k+1}$, $K = 8$, $\beta = 2$, and $\eta = 0.1$. Because $\beta > 1$ the update extrapolates toward the matched statistics; applying it at every step makes alignment part of the dynamics rather than a post-process, and it supplies HH statistics without a learned head.

E. Metric-free Stopping

Training regresses a small latent field, needs no decoding, inversion, or adversary, and runs 52 steps per epoch at batch 96. After each epoch we probe a fixed batch of four latents at $t = 0.5$: ρ_e is the ratio of LL to HF gradient norm on the shared trunk. Training stops at the first epoch where the target-route gate grows and $\rho_e/\rho_{e-1} \leq 0.65$, i.e., when high-frequency learning takes over the trunk. On D5 this fires at epoch 4 (82.8 s), which is also the DINO-S peak of the 15-epoch curve; no external metric is used online.

IV. EXPERIMENTS

A. Setup

Benchmarks. *D5-512*: five WikiArt styles (Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e) at 512 px, 3,600 training images each; 30 test sources per style are sent to all five targets (750 ordered requests, same-style rows kept). *P2A-256*: a 256 px grid over photo, Monet, Van Gogh, Cézanne, and Hayao. *R5-WikiArt*: five randomly drawn WikiArt families (Cubism, Expressionism, Pop Art, Romanticism, Symbolism) at 512 px.

Metrics. DINO-S (primary) is the maximum cosine between the DINOv2-small CLS feature [38] of the output and held-out references of the target style; CLIP-S is CLIP ViT-B/32 [39] similarity to the target-style prototype; DINO-C is DINOv2 similarity to the source; LPIPS uses AlexNet. IDT and TGT (a fixed held-out reference of the target style) are computed per board, and we inspect $M_{\text{style}}(y) > M_{\text{style}}(y_{\text{IDT}})$ and $\text{LPIPS}(y) < \text{LPIPS}(y_{\text{TGT}})$, $\text{DINO} - C(y) > \text{DINO} - C(y_{\text{TGT}})$ separately rather than combining them into one score.

Baselines and implementation. We compare training-free diffusion editors (SD-Turbo [40], StyleAligned, Z-STAR, StyleShot, StyleID), learned models (CUT, SaMST, SaMam, StyTR-2, AesPA-Net), the Seedream 4.5 API [41], and an analytic Latent-WCT, each at its released or recommended operating point. WEAVE uses AdamW ($\text{lr } 2 \times 10^{-4}$, cosine), bf16, and one RTX 3060 (12 GB); training time excludes one-off latent caching. Inference time covers the 8-step ODE and VAE decoding for 750 images.

B. Main Results

Auditing the field. The last column of Table I counts the benchmarks on which a method stays inside the sandwich; only five of thirteen reach 3/3. The failures are instructive. SD-Turbo attains the best D5 LPIPS (0.003) precisely because it barely edits the input: its CLIP-S equals that of IDT. SaMam, the strongest compact baseline on content, has CLIP-S below

TABLE I

MAIN COMPARISON (D-S: DINO-S, C-S: CLIP-S, LP: LPIPS, D-C: DINO-C). EXCLUDING IDT AND TGT, THE BEST ENTRY PER COLUMN IS **BOLD** AND THE SECOND UNDERLINED. †: STYLE NOT ABOVE IDT; ‡: CONTENT BEYOND TGT. PARAMS IN MILLIONS WITH FROZEN PARAMETERS AS SUPERScript; PARAMS AND TRAIN RANK TRAINED METHODS. WEAVE USES ITS SELECTED D5 CHECKPOINT ON D5 AND DATASET-SPECIFIC CHECKPOINTS ON P2A/R5; COST REFERS TO THE D5 MODEL. VALID: NUMBER OF BENCHMARKS WITHOUT † OR ‡. TIMES ON ONE RTX 3060; *: AUTHOR REPORT ON FOUR GPUS [16]; ~3 h: Z-STAR ESTIMATED FROM STEP-SCALED RUNS (FULL 512 PX RUNS EXCEED 12 GB).

Method	D5-512				P2A-256				R5-WikiArt				Params	Train	Infer	Valid
	D-S†	C-S†	LP↓	D-C†	D-S†	C-S†	LP↓	D-C†	D-S†	C-S†	LP↓	D-C†	(M)	(min)	(750)	boards
IDT (identity)	0.419	0.693	0.000	1.000	0.416	0.663	0.000	1.000	0.433	0.731	0.000	1.000	0	free	0 s	—
TGT (target style)	1.000	0.863	0.776	0.215	1.000	0.835	0.800	0.169	1.000	0.796	0.747	0.230	0	free	0 s	—
SD-Turbo	0.484	0.693 [†]	0.003	0.922	0.341 [†]	0.674	0.603	0.279	0.505	0.767	0.449	0.538	0 ⁺¹¹⁰⁰	free	5 m	1/3
StyleAligned	0.675	0.780	0.869 [‡]	0.239	0.612	0.768	0.786	0.310	0.649	0.824	0.829 [‡]	0.315	0 ⁺¹¹⁰⁰	free	77 m	1/3
Z-STAR	0.449	0.784	0.347	0.549	0.498	0.786	0.332	0.552	0.514	<u>0.822</u>	0.384	0.526	0 ⁺¹¹⁰⁰	free	~3 h	3/3
StyleShot	<u>0.563</u>	<u>0.787</u>	0.765	0.377	<u>0.552</u>	<u>0.774</u>	0.713	0.335	0.484	0.787	0.795 [‡]	0.380	0 ⁺²⁴⁰⁰	free	5.1 h	2/3
StyleID	0.548	0.822	0.552	0.416	0.256 [†]	0.648 [†]	0.691	0.147 [‡]	<u>0.551</u>	0.815	0.493	0.385	0 ⁺¹¹⁰⁰	free	63 m	2/3
CUT	0.471	0.714	0.374	0.795	0.539	0.754	0.494	0.702	0.441	0.710 [†]	0.620	<u>0.795</u>	<u>7.0</u>	322.6	5 m	2/3
SaMST	0.271 [†]	0.618 [†]	0.749	0.145 [‡]	0.501	0.709	0.398	0.838	0.461	0.667 [†]	0.612	0.615	8.3	<u>39.5</u>	10 m	1/3
SaMam	0.477	0.582 [†]	<u>0.243</u>	<u>0.812</u>	0.505	0.677	0.205	0.870	0.503	0.712 [†]	0.227	0.925	8.8	436.0	17.6 m	1/3
StyTR-2	0.501	0.709	<u>0.596</u>	<u>0.747</u>	0.515	0.722	0.495	0.555	0.533	0.759	0.553	0.737	35.4 ^{+12.9}	~1440*	38 m	3/3
AesPA-Net	0.520	0.724	0.548	0.601	0.498	0.744	0.571	0.522	0.545	0.782	0.409	0.728	11.3 ^{+20.0}	—	23 m	3/3
Seedream 4.5	0.486	0.720	0.477	0.739	0.517	0.752	0.227	0.785	0.504	0.742	0.486	0.725	—	API	—	3/3
Latent-WCT	0.362 [†]	0.673 [†]	0.441	0.559	0.338 [†]	0.738	0.585	0.386	0.414 [†]	0.710 [†]	0.444	0.553	0	free	18 s	0/3
WEAVE (ours)	0.492	0.713	0.260	0.810	0.480	0.668	0.312	<u>0.861</u>	0.523	0.775	<u>0.289</u>	0.772	1.04 ⁺⁸⁴	1.4	<u>126 s</u>	3/3

IDT on two benchmarks, so its low LPIPS partly reflects the identity shortcut. At the other extreme, StyleAligned has the highest DINO-S everywhere but moves farther from the source than an unrelated target painting on D5 and R5. Latent-WCT never clears IDT, showing that wavelet statistics alone do not produce transfer; the learned transport is necessary.

Comparison among valid methods. Restricted to the five methods that pass everywhere (Z-STAR, StyTR-2, AesPA-Net, Seedream 4.5, and WEAVE), WEAVE has the highest DINO-C on all three benchmarks (0.810/0.861/0.772) and the lowest LPIPS on D5 and R5, while its DINO-S lies within the range of the others. On D5 it reaches higher DINO-S than the commercial Seedream 4.5 at about half the LPIPS (0.260 vs. 0.477). The stronger D5-512 profile is consistent with native high-frequency coefficients preserving detail that is compressed at 256 px.

Cost. WEAVE has the fewest trainable parameters and the shortest training of all learned methods (1.4 min vs. 39.5–1440 min, 28–1000× less) and needs 168 ms per 512 px image including VAE decoding. Among the valid methods, the next cheapest learned model (AesPA-Net) has 11× more trainable parameters and 11× longer inference.

Qualitative comparison. Fig. 4 shows the same pattern visually: editors that score high on style redraw the scene, compact baselines add color casts but little texture, and WEAVE transfers brushwork onto the preserved source layout.

C. Mechanistic Ablations

Table II changes one component at a time on the same 750 requests. Inference-time controls reuse the selected checkpoint; retrained variants use seed 42 and are reported at their best-DINO-S epoch. Removing stepwise AdaIN lowers DINO-

TABLE II
COMPONENT ABLATION ON D5-512 (750 REQUESTS).

Variant	DINO-S†	CLIP-S†	LPIPS↓	DINO-C†
WEAVE (epoch 4)	0.4918	0.7128	0.2595	0.8102
w/o stepwise AdaIN ($\beta=0$)	0.4821	0.7005	0.2258	0.8455
w/o HF conditioning	0.4825	0.7157	0.2837	0.8125
direct endpoint ($z_1^* = z_s$)	0.4894	0.7172	0.3138	0.7966
$\lambda_{LL}=1.0$	0.4910	0.7180	0.2626	0.7874
learned HH head	0.4930	0.7164	0.2670	0.8061

S by 0.0097 and CLIP-S by 0.0123 and moves the output toward the source. Removing the high-frequency conditioning lowers DINO-S and raises LPIPS. Replacing the source-aligned endpoint with the direct target has the largest content cost (LPIPS 0.2595→0.3138), and its peak arrives nine epochs later. Raising λ_{LL} from 0.3 to 1.0 changes peak DINO-S by less than 0.001 but lowers DINO-C to 0.787, so the method does not hinge on a finely tuned weight. A learned HH head gives a small style gain for a small content cost, so we keep the simpler model.

D. Robustness and Transfer

Seeds and stopping. The stopping rule selects epochs 4, 4, and 3 for seeds 42, 7, and 123, giving DINO-S 0.4918, 0.4910, and 0.4862 (LPIPS 0.260, 0.267, 0.255). For seeds 42 and 7, where full 15-epoch curves exist, the selected epoch coincides with the DINO-S peak.

Reference pools. Resampling a size- m reference pool per target style 1,000 times, the board-mean DINO-S margin of WEAVE over IDT is positive in every draw: 0.030 (95% interval [0.020, 0.039]) for $m = 8$ and 0.032 ([0.023, 0.038])

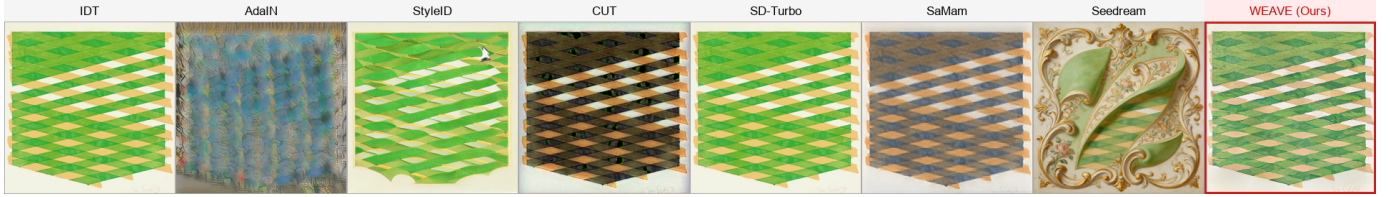


Fig. 4. Extreme cross-domain request (Minimalism $\rightarrow$ Rococo). Diffusion editors substantially alter source structure, and several lightweight or classical outputs remain close to identity. WEAVE retains the source layout while adding the painterly target appearance.

TABLE III
TRANSFER BEYOND THE TRAINING SETTING. TOP: ZERO-SHOT ON FIVE UNSEEN WIKIART FAMILIES (750 PAIRS, D5 CHECKPOINTS). BOTTOM: HAAR ALIGNMENT INSIDE A FROZEN SD1.5 EDITOR (600 PAIRS).

Method	DINO-S $\uparrow$	CLIP-S $\uparrow$	LPIPS $\downarrow$	DINO-C $\uparrow$
SaMST	0.3913	0.7485	0.6213	0.2680
SaMam	0.5466	0.7577	0.3990	0.7369
WEAVE	0.5313	0.7742	0.2802	0.7587
SD1.5 img2img	0.3974	0.7221	0.3591	0.6589
+ Haar alignment	0.4080	0.7366	0.3630	0.6531

for $m = 16$. About 69% of individual requests improve, so the claim is board-level rather than per image.

Unseen families. Applied without retraining to five held-out WikiArt families (Table III, top), the D5 checkpoint gives the best CLIP-S, LPIPS, and DINO-C; SaMam is 0.015 higher on DINO-S at a much higher LPIPS.

Plug-in for a frozen editor. We add the same Haar-domain alignment (LH/HL/HH, scale 0.8, LL untouched) to a frozen SD1.5 image-to-image pipeline (empty prompt, strength 0.5, 20 DDIM steps). On 600 cross-style D5 pairs, DINO-S and CLIP-S rise significantly (paired t -test, $p < 10^{-15}$) at a small LPIPS change (Table III, bottom), for 4.9 ms per image and no weight update.

V. LIMITATIONS AND CONCLUSION

WEAVE is evaluated with one SD1.5 VAE and a single Haar level; styles dominated by global palette or layout, or expressed across several scales, may need a deeper decomposition. Haar separates latent coordinates, not semantic factors, and the nonlinear decoder can couple them. The stopping rule is validated on one architecture and three seeds, and IDT/TGT are board-specific references measured with imperfect perceptual proxies. Baselines use their recommended operating points rather than a compute-matched search.

The identity shortcut is easy to miss because the metrics that should catch it reward it. Two trivial references, IDT and TGT, make it visible in raw metric units, and under them most of the field fails somewhere. The shortcut has a spectral cause: coarse structure dominates the gradient while carrying little style. WEAVE addresses that cause directly, anchoring structure at the source, learning texture transport in explicit subbands, and injecting high-frequency statistics along the trajectory. The result is the most content-preserving valid transfer we measured, from a 1.04M-parameter model

trained in 1.4 minutes on a consumer GPU, which makes style transfer practical for resource-constrained multimedia applications. Code, checkpoints, and the 750-request protocol will be released; the supplement details the protocol, per-epoch curves, and additional controls.

REFERENCES

- [1] Chenfei Wu, Jiahao Li, Jingren Zhou, et al., “Qwen-image technical report,” arXiv preprint arXiv:2508.02324, 2025.
- [2] Black Forest Labs, “FLUX.2: Frontier Visual Intelligence,” <https://bfl.ai/blog/flux-2>, 2025.
- [3] Hongda Liu, Longguang Wang, Weijun Guan, Ye Zhang, and Yulan Guo, “Pluggable style representation learning for multi-style transfer,” in *Proceedings of the Asian Conference on Computer Vision*, 2024.
- [4] Hongda Liu, Longguang Wang, Ye Zhang, Ziru Yu, and Yulan Guo, “Samam: Style-aware state space model for arbitrary image style transfer,” arXiv preprint arXiv:2503.15934, 2025.
- [5] Taesung Park, Alexei A. Efros, Richard Zhang, and Jun-Yan Zhu, “Contrastive learning for unpaired image-to-image translation,” in *Proceedings of the European Conference on Computer Vision*, 2020.
- [6] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang, “The unreasonable effectiveness of deep features as a perceptual metric,” in *CVPR*, 2018.
- [7] Matthias Wright and Björn Ommer, “Artfid: Quantitative evaluation of neural style transfer,” arXiv preprint arXiv:2207.12280, 2022.
- [8] Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Li, Fred Hamprecht, Yoshua Bengio, and Aaron Courville, “On the spectral bias of neural networks,” in *ICML*, 2019.
- [9] Zhi-Qin John Xu, Yaoyu Zhang, and Tao Luo, “Overview frequency principle/spectral bias in deep learning,” arXiv preprint arXiv:2201.07395, 2022.
- [10] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer, “High-resolution image synthesis with latent diffusion models,” in *CVPR*, 2022.
- [11] Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge, “A neural algorithm of artistic style,” in arXiv preprint arXiv:1508.06576, 2015.
- [12] Leon A. Gatys, Alexander S. Ecker, and Matthias Bethge, “Image style transfer using convolutional neural networks,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016.
- [13] Xun Huang and Serge Belongie, “Arbitrary style transfer in real-time with adaptive instance normalization,” in *ICCV*, 2017.
- [14] Yijun Li, Chen Fang, Jimei Yang, Zhaowen Wang, Xin Lu, and Ming-Hsuan Yang, “Universal style transfer via feature transforms,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017.
- [15] Dae Young Park and Kwang Hee Lee, “Arbitrary style transfer with style-attentional networks,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.
- [16] Yingying Deng, Fan Tang, Weiming Dong, Wen Sun, Feiyue Huang, and Changsheng Xu, “Stytr2: Image style transfer with transformers,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022.
- [17] Kibeom Hong, Sangryul Jeon, Hyesong Yang, Jianlong Fu, and Hyeran Byun, “Aespa-net: Aesthetic pattern-aware style transfer networks,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.
- [18] Albert Gu and Tri Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” arXiv preprint arXiv:2312.00752, 2023.

- [19] Chenlin Meng, Yutong He, Yang Song, Jiaming Song, James Blackwell, Jiazhao Jia, and Stefano Ermon, “Sdedit: Guided image synthesis and editing with stochastic differential equations,” in *Proceedings of the IEEE International Conference on Computer Vision*, 2021.
- [20] Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or, “Prompt-to-prompt image editing with cross attention control,” *arXiv preprint arXiv:2208.01626*, 2022.
- [21] Jiwoo Chung, Sangeek Hyun, and Jae-Pil Heo, “Style injection in diffusion: A training-free approach for adapting large-scale diffusion models for style transfer,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [22] Amir Hertz, Andrey Voynov, Shlomi Fruchter, and Daniel Cohen-Or, “Style aligned image generation via shared attention,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024.
- [23] Yuxin Zhang, Nisha Huang, Fan Tang, Haibin Huang, Chongyang Ma, Weiming Dong, and Changsheng Xu, “Inversion-based style transfer with diffusion models,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- [24] Zhi-Song Lu, Sai Vukoti, Isma Hadji, W. Dorken, and Guy Levi, “Z-star: Training-free zero-shot style transfer via attention rearrangement,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.
- [25] Hu Ye, Jun Zhang, Sibio Liu, Xiao Han, and Wei Yang, “Ip-adapter: Text compatible image prompt adapter for text-to-image diffusion models,” *arXiv preprint arXiv:2308.06721*, 2023.
- [26] Junyao Gao, Yanchen Liu, Yanan Sun, Yinhao Tang, Yanhong Zeng, Kai Chen, and Cairong Zhao, “Styleshot: A snapshot on any style,” *arXiv preprint arXiv:2407.01414*, 2024.
- [27] Xingchao Liu, Chengyue Gong, and Qiang Liu, “Flow straight and fast: Learning to generate and transfer data with rectified flow,” in *International Conference on Learning Representations*, 2023.
- [28] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le, “Flow matching for generative modeling,” *arXiv preprint arXiv:2210.02747*, 2022.
- [29] Valentin De Bortoli, James Thornton, Jeremy Heng, and Arnaud Doucet, “Diffusion schrödinger bridge with applications to score-based generative modeling,” *arXiv preprint arXiv:2106.01357*, 2021.
- [30] Yuyang Shi, Valentin De Bortoli, Andrew Campbell, and Arnaud Doucet, “Diffusion schrödinger bridge matching,” in *Advances in Neural Information Processing Systems*, 2023.
- [31] Guan-Hong Liu, Arash Vahdat, De-An Huang, Evangelos A. Theodorou, Weili Nie, and Anima Anandkumar, “I2sb: Image-to-image schrödinger bridge,” in *International Conference on Machine Learning*, 2023.
- [32] Xuan Su, Jiaming Song, Chenlin Meng, and Stefano Ermon, “Dual diffusion implicit bridges for image-to-image translation,” in *International Conference on Learning Representations*, 2023.
- [33] Mu Cai, Hong Zhang, Huijuan Huang, Qichuan Geng, Yixuan Li, and Gao Huang, “Frequency domain image translation: More photo-realistic, better identity-preserving,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021.
- [34] Sucheng Ren, Qihang Yu, Ju He, Xiaohui Shen, Alan Yuille, and Liang-Chieh Chen, “Frequency-aware flow matching for high-quality image generation,” *arXiv preprint arXiv:2604.15521*, 2026.
- [35] Hao Phung, Quan Dao, and Anh Tran, “Wavelet diffusion models are fast and scalable image generators,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- [36] Jaejun Yoo, Youngjung Uh, Sanghyuk Chun, Byeongkyu Kang, and Jung-Woo Ha, “Photorealistic style transfer via wavelet transforms,” in *ICCV*, 2019.
- [37] Ingrid Daubechies, “Ten lectures on wavelets,” *SIAM*, 1992.
- [38] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al., “DINOv2: Learning robust visual features without supervision,” *Transactions on Machine Learning Research*, 2024.
- [39] Alec Radford, Jong Wook Kim, Chris Hallacy, et al., “Learning transferable visual models from natural language supervision,” in *International Conference on Machine Learning (ICML)*, 2021, pp. 8748–8763.
- [40] Axel Sauer, Dominik Lorenz, Andreas Blattmann, and Robin Rombach, “Adversarial diffusion distillation,” *arXiv preprint arXiv:2311.17042*, 2023.
- [41] ByteDance Seed, “Seedream 4.0: Toward next-generation multimodal image generation,” *arXiv preprint arXiv:2509.20427*, 2025.